



Design and optimization of discrete-time delta-sigma modulators

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ARTICLE INFO

Keywords:

Dynamic range
Delta-sigma modulator (DSM)
Discrete-time
Quantizer

ABSTRACT

—For high-precision sensor applications, high dynamic range discrete-time delta-sigma modulators (DSMs) are key components. For the design of high dynamic range DSMs, the system architecture is crucial. The system requires strong in-band noise suppression for small input amplitudes and a large maximum stable amplitude (MSA) for large input amplitudes. This article presents the system design and optimization method of discrete-time delta-sigma modulator. It provides a detailed discussion on the behavior-level modeling and characteristics of low-bit and multi-bit quantizers. Additionally, design considerations of quantizers in DSM systems are analyzed. An approach for optimizing DSM systems by leveraging the properties of quantizers is proposed, along with a simple optimization case study. Potential future application scenarios are also presented. The purpose of this article is to offer design guidelines for developing well-performing DSMs systems.

1. Introduction

Sensors are among the most widely used devices, covering fields such as measurement, biomedical, imaging, and audio applications. Analog-to-digital converters (ADCs), as critical components of sensors, play a key role, with the dynamic range of an ADC being a key determinant of sensor performance. In response to market demands, high-precision sensors typically require both a high dynamic range and low power to meet specific application requirements. Successive-Approximation Register (SAR) ADCs offer high energy efficiency at moderate speeds and resolutions [1,2]. However, although SAR ADCs can achieve resolutions greater than 12 bits, this comes at the cost of complex calibration techniques and large input sampling capacitors [3–7]. In contrast, delta-sigma modulators offer architectural advantages in low-bandwidth, high dynamic range designs, making them an ideal choice for high-precision sensor applications [8–14].

High dynamic range can be achieved by DSMs with higher loop gain, higher OSR, and multi-bit quantizers. However, higher loop gain comes at the expense of the system's MSA, higher OSR comes at the cost of increasing both analog and digital power consumption, and multi-bit quantizers require additional mismatch error shaping overhead [15, 16]. Therefore, a good system design requires trade-offs between loop gain, OSR, and quantizer bits. The equivalent model of the delta-sigma system is shown in Fig. 1, which assumes that the quantizer has a fixed gain factor k . k directly affects the loop gain and MSA of the system. Therefore, k has the right to vote for good or bad system

performance. Conventional multi-bit quantizers usually contribute a gain factor k close to 1, which makes the behavior of the system easier to capture

in the mathematical domain. However, the equivalent gain of a low-bit quantizer usually varies with the input, which makes the system behavior difficult to predict.

This article presents the system design and optimization method of discrete-time delta-sigma modulator. The behavior-level modeling of quantizers is discussed in detail, including the empirical prediction method for equivalent gain, characteristics of mid-tread and mid-rise quantizers, as well as the relationship between input amplitude, threshold voltage, and equivalent gain. The behavior of the quantizer in a DSM system is also examined, covering the linear model characteristics of the quantizer in DSM, nonlinear considerations in DSM, limitations of empirical evaluation methods, and linearity considerations of feedback DACs. Additionally, how to flexibly utilize the gain of the quantizer to achieve system optimization is discussed, along with a simple case study.

The organization of the remaining sections of this article is as follows. Section 2 analyzes the modeling of low-bit and multi-bit quantizers. Section 3 presents the design considerations of the quantizer in DSM systems. Section 4 shows the potential of using quantizers to optimize DSM system design. Section 5 gives a summary of the article.

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<https://doi.org/10.1016/j.mejo.2024.106529>

Received 23 September 2024; Received in revised form 19 November 2024; Accepted 9 December 2024

Available online 10 December 2024

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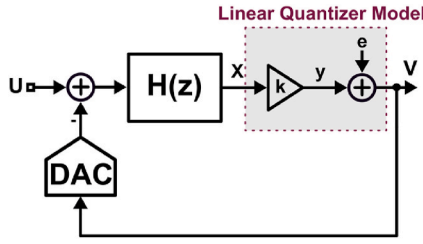


Fig. 1. The equivalent model of the DSM system.

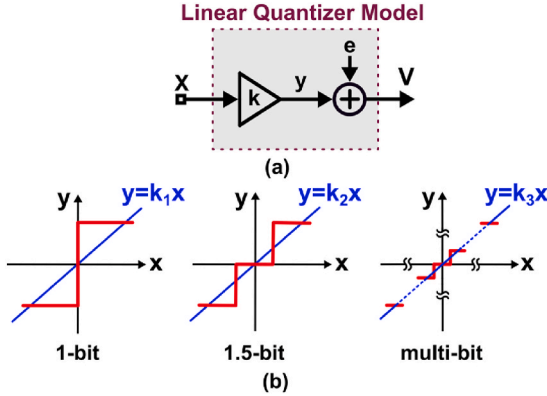


Fig. 2. The equivalent gain model of a quantizer.

2. Modeling of quantizers

Quantization is a nonlinear and memoryless operation. Based on the characteristics of the zero level, quantizers are generally classified into mid-rise quantizers and mid-tread quantizers. To simplify the analysis, it is assumed that the equivalent gain k of the quantizer is linear. Therefore, the linear model of the quantizer can be expressed as $v = y + e = kx + ev$ as shown in Fig. 2(a). To make the analysis more intuitive, only 1-bit and 1.5-bit quantizers, including both mid-rise and mid-tread types, are considered for the analysis of low-bit quantizers. The equivalent gain k models for 1-bit, 1.5-bit, and multi-bit quantizers are shown in Fig. 2(b).

In a multi-bit quantizer, an intuitive assumption is that k_3 is fixed, or that the value of k_3 is around 1, which seems reasonable. However, in a 1-bit quantizer, there are only two quantization levels, and multiple k_1 lines can satisfy the maximum quantization error of Δ , giving the impression that the equivalent gain has infinite possibilities, making the behavior of the quantizer difficult to predict, which is not ideal for design purposes. Similarly, the 1.5-bit quantizer presents similar challenges.

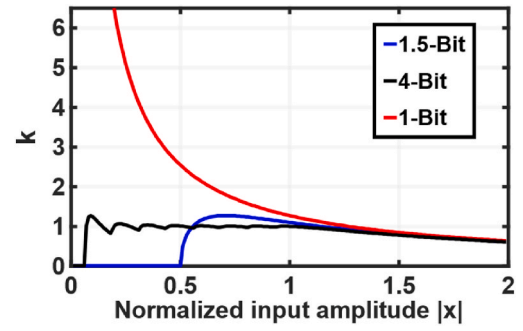
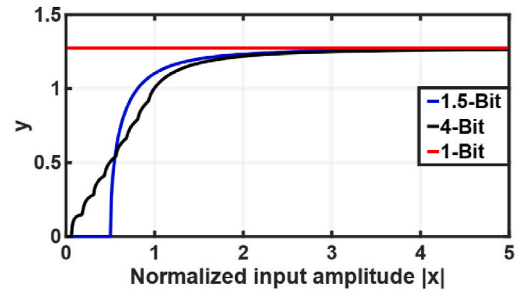
Therefore, it is necessary to determine the equivalent gain of the quantizer. One empirical evaluation method defines the equivalent gain k of the quantizer as the value that minimizes the mean squared error between v and y [17]. The formula is:

$$\sigma_e^2 = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^N e^2[n] = \lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^N (V[n] - kx[n])^2 \quad (1)$$

Therefore, the equivalent gain k of the quantizer can be expressed as:

$$k = \frac{\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^N (V[n]x[n])}{\lim_{N \rightarrow \infty} \frac{1}{N} \sum_{n=0}^N (x[n]x[n])} \quad (2)$$

Based on the above formula, the equivalent gain k of the quantizer can be well-fitted. It is important to note that this is a fitting model derived from a large amount of data in the time domain, and the fitting

Fig. 3. Simulated the behavior of the equivalent gain k of the quantizer at different normalized input amplitudes $|x|$.Fig. 4. Simulated the behavior of y at different normalized input amplitudes $|x|$.

results are strongly correlated with the properties of the input signal. This also implies that this empirical fitting method has a certain level of validity.

For a sinusoidal input with an amplitude of $|x|$, based on the previous analysis, the equivalent gain k of the quantizer can be obtained, as shown in Fig. 3. Using the relation $V = kx + e = y + e$, the behavior curves of y for different normalized input amplitudes $|x|$ can be plotted, as

shown in Fig. 4. The behavior in Fig. 4 can be predicted from Fig. 3. The gain k of the 1-bit quantizer increases continuously as $|x|$ decreases. Theoretically, it can become infinitely large. As a result, the amplitude of y in a 1-bit quantizer remains almost constant, which means that the modulator has a higher gain at small input amplitudes. Traditional design approaches tend to focus more on the variation of k , while overlooking the fact that y remains constant. In mathematical computations, the constancy of y simplifies the design of the NTF in DSM.

To simplify the analysis, a multi-bit quantizer is exemplified by a 4-bit quantizer. When the normalized input amplitude $|x|$ exceeds the threshold V_{TH} , the gain k remains close to 1. This is interesting because the gain k of a multi-bit quantizer does not change with the input amplitude, making its behavior easier to predict. However, near the left side of V_{TH} , the gain k suddenly drops, a problem observed in mid-tread quantizers.

The equivalent gain k of a 1.5-bit quantizer appears to be a compromise between the 1-bit and multi-bit quantizers [18,19]. In the right half of the curve, as $|x|$ decreases, k gradually increases, resembling the behavior of a 1-bit quantizer. An interesting phenomenon occurs when $|x|$ approaches V_{TH} , where k experiences a sharp drop. This behavior is not surprising since the 1.5-bit quantizer is a low-bit mid-tread quantizer.

According to the equation $V = kx + e$, the actual input of the quantizer is given by $y = kx$. To better study the effect of V_{TH} on the equivalent gain k of the quantizer, we assume a quantization scenario where $|x| = aV_{TH}$. Assuming that when $V_{TH} = 0.5$, the corresponding quantizer equivalent gain is represented by the curve $k_{V_{TH}=0.5}$. Therefore,

under the same quantization scenario, the equivalent gain k for different V_{TH} values can be expressed as:

$$k = \frac{0.5}{V_{TH}} k_{V_{TH}=0.5} \quad (3)$$

According to Equation (3), the next state of k can be proportionally adjusted by tuning V_{TH} . This means that for different inputs $|x|$, as long as an appropriate V_{TH} is found, the value of k in the system will remain unchanged. This is meaningful.

According to $y = kx$, the value of y can be calculated as:

$$y = 0.5 \bullet a \bullet k_{V_{TH}=0.5} \quad (4)$$

$k_{V_{TH}=0.5}$ is a function dependent on a . From equation (4), we know that as $|x|$ changes, dynamically adjusting V_{TH} can maintain the same quantization conditions. This ensures that the equivalent gain of the 1.5-bit quantizer will automatically adjust to keep the amplitude of y constant. Compared to Equation (3), the design guidance based on y provided by Equation (4) is easier to observe.

3. Analysis of quantizer in DSM system

The quantizer is one of the key modules of DSM. Previous analyses were based on linear assumptions. Although the non-ideal factors of the quantizer can be suppressed by the loop filter, there are some situations that need to be considered. Next, we will discuss the characteristics of linear quantizer models in DSM, the characteristics of nonlinear quantizer models in DSM, the limitations of empirical evaluation methods, and the linearity of feedback DACs.

3.1. The characteristics of linear quantizer models in DSM

Based on the analysis in Section 2, integrating the quantizer into the DSM system will make the behavior of the DSM more predictable.

According to Fig. 3, the gain k of the 1-bit quantizer increases continuously as $|x|$ decreases. Theoretically, it can become infinitely large, meaning that the modulator can achieve higher loop gain at small input amplitudes, resulting in stronger noise-shaping capabilities. However, for high-order single-loop 1-bit delta-sigma modulators, they typically have smaller MSA.

For multi-bit quantizers, when the normalized input amplitude $|x|$ exceeds V_{TH} , k remains close to 1. This is interesting because the noise transfer function of multi-bit quantizers does not vary with changes in input amplitude, making the system performance easier to predict. The equivalent gain k of the 1.5-bit quantizer appears to represent a trade-off between 1-bit and multi-bit quantizers. In the right half of the curve, as $|x|$ decreases, k gradually increases, resembling the behavior of a 1-bit quantizer. An intriguing phenomenon is the sharp drop in k as $|x|$ approaches V_{TH} . This is undesirable since such behavior limits the input dynamic range of the delta-sigma modulator.

Fortunately, Equation (4) indicates that at small input amplitudes, the quantization conditions can be adjusted by varying V_{TH} , thereby improving the dynamic range limitations of a delta-sigma modulator based on a 1.5-bit quantizer. Additionally, the y model analysis presented in Fig. 4 is beneficial for the design of DSMs, as it makes the complex behavior of quantizers more predictable.

3.2. The characteristics of nonlinear quantizer models in DSM

Generally, multi-bit quantizers exhibit lower nonlinearity, and due to the suppression effect of the loop filter, the nonlinearity introduced by the multi-bit quantizer can often be disregarded. However, the nonlinearity of low-bit quantizers, particularly the 1.5-bit quantizer, is relatively poor. The harmonic distortion energy caused by the quantizer's nonlinearity can reach levels comparable to in-band quantization noise. Additionally, when harmonics are located at the bandwidth edges, the

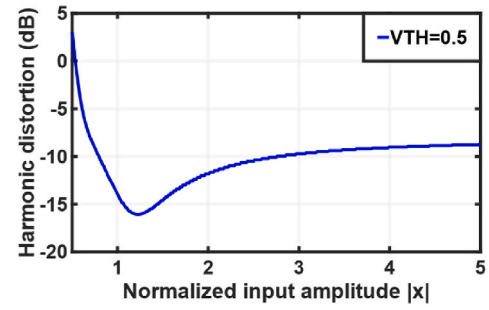


Fig. 5. The relationship between normalized harmonic distortion and input amplitude $|x|$ in a 1.5-bit quantizer.

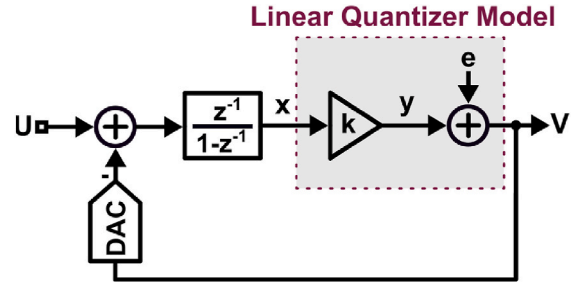


Fig. 6. The equivalent model of the first-order DSM system.

suppression effect of the loop filter on these harmonics is weakest. Therefore, the nonlinearity of the 1.5-bit quantizer must be taken into account. The actual k -model of the 1.5-bit quantizer can be expressed as:

$$y = k_0 + k_1 x^1 + k_2 x^2 + \dots + k_{n-1} x^{n-1} + k_n x^n \quad (5)$$

To simplify the analysis, we take $V_{TH} = 0.5$ as an example and use a sine wave with an input amplitude of $|x|$ to stimulate the 1.5-bit quantizer. The resulting normalized harmonic distortion energy curve of the quantizer output bitstream is shown in Fig. 5. Observing the curve, we find that the harmonic distortion energy of the 1.5-bit quantizer initially decreases as $|x|$ increases, then begins to increase. The distortion is minimized when the input amplitude $|x|$ is approximately 2.4 times V_{TH} . Nonlinear analysis is important, as Fig. 5, together with Fig. 3, can quickly find the optimal k -value for the best system linearity. Compared to traditional design methods based on exhaustive search [17], it saves a significant amount of time.

Although the linearity of the 1.5-bit quantizer is relatively poor, it is attenuated by the suppression of the loop filter. In a practical delta-sigma modulator, the equivalent gain k of the quantizer has different contributions to signal and noise, so the assessment of nonlinearity in the 1.5-bit quantizer must be placed within the context of a specific modulator architecture. Generally, under overload conditions, the k for noise is less than that for the input signal. Fortunately, the harmonic distortion energy of the 1.5-bit quantizer is related to V_{TH} , allowing us to minimize distortion in the delta-sigma modulator system by selecting an appropriate V_{TH} .

3.3. The limitations of empirical evaluation methods

In the previous analysis, the effective gain k of the quantizer can be empirically assessed by separately applying a sine wave to the quantizer, where the equivalent gain k is defined by minimizing the mean square error between V and y . However, analyzing the equivalent gain k of the quantizer independently from the DSM system is not rigorous, as the quantizer input $|x|$ does not strictly follow the input sine wave and instead contains noisy components. Therefore, evaluating the equivalent gain k of the quantizer in the system requires a time-domain analysis.

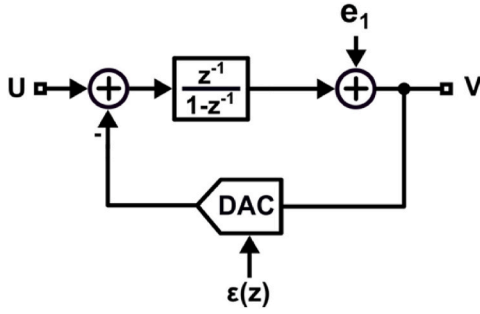


Fig. 7. First-order delta-sigma ADC model with feedback DAC mismatch error.

To simplify the analysis, we assume that the contribution of the quantizer's equivalent gain k to the input signal and noise of the delta-sigma ADC is the same. For further simplification, let's take the simple DSM shown in Fig. 6 as an example to demonstrate how to empirically evaluate the quantizer's equivalent gain k ; similar derivations can be applied to other DSM architectures. Assuming a DAC gain of 1, after M cycles, the time-domain expression for x is:

$$x[M] = \sum_{j=0}^{M-1} (U[j] - V[j]) \quad (6)$$

Based on Equation (6), the equivalent gain k of the quantizer in a DSM can be well-fitted. This is meaningful because it allows for monitoring changes in the quantizer's equivalent gain k within the DSM system at different input amplitudes. In traditional manual calculations, we often assume $k = 1$, which makes the model less accurate. The k model provides a more intuitive approach to designing and analyzing the loop filter, making the process clearer and more accurate for system optimization.

In practice, since k contributes differently to noise, sinusoidal signals, and other noisy components, the empirical evaluation method still has some inaccuracies.

3.4. The linearity of feedback DACs

Fig. 7 shows the first-order delta-sigma modulator model with feedback DAC mismatch error, whose output V is expressed as:

$$V(z) = z^{-1} \bullet U(z) + z^{-1} \bullet \epsilon(z) - (1 - z^{-1}) \bullet e_1(z) \quad (7)$$

It shows that the nonlinearity of the feedback DAC directly overlays on the input signal, which is quite detrimental. For a 1-bit quantizer, there are only two quantization levels and feedback DAC levels, resulting in absolutely linear feedback. However, the limitation of the 1-bit quantizer is large quantization error, meaning that improving the peak signal-to-quantization noise ratio (SQNR) of the delta-sigma modulator can only be achieved through higher OSR and loop order, which are constrained by power consumption and system stability.

Multi-bit quantizers significantly reduce the system's quantization error, relieving the swing stress on the integrator and improving the system's MSA. However, the feedback DAC of multi-bit quantizers requires multiple feedback levels, and the nonlinearity caused by mismatches between feedback units directly overlays on the input signal, degrading the signal-to-noise and distortion ratio (SNDR) performance of the delta-sigma modulator. Therefore, the feedback DAC units of multi-bit quantizers often require mismatch error calibration techniques. Common calibration techniques include dynamic element matching (DEM), which primarily consists of random selection (RS) and data-weighted averaging (DWA) [20–22]. In some high-SNDR application scenarios, higher-order mismatch noise shaping techniques may also be necessary [23,24].

In a 1.5-bit quantizer, a tri-level feedback CDAC is typically used to mitigate the linear effects introduced by capacitor mismatches. For area

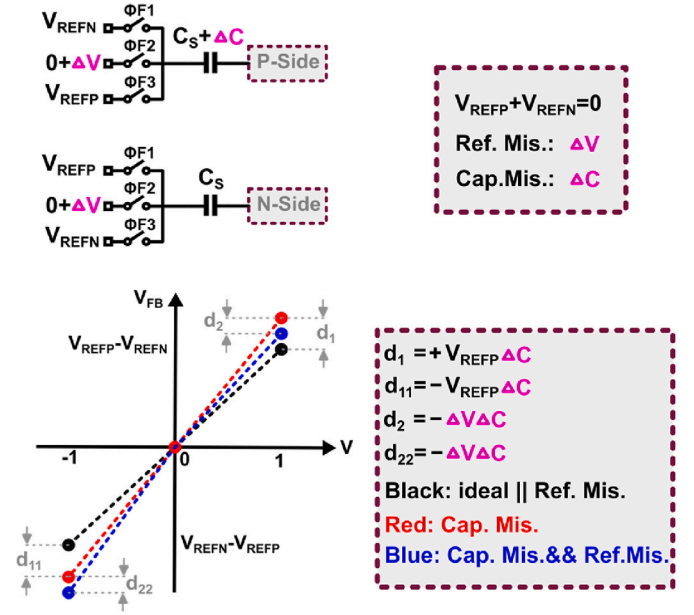


Fig. 8. The differential behavior of the tri-level DAC (top) and the feedback transfer curve caused by capacitor and reference voltage mismatch (bottom).

and energy efficiency considerations, it is appropriate to reuse the input sampling capacitors from the first-stage integrator for feedback. The differential behavior of the tri-level DAC and the feedback transfer curve caused by capacitor and reference voltage mismatches are shown in Fig. 8. Ideally, the behavior of the feedback voltage V_{FB} can be represented as:

$$V_{FB,ideal} = \begin{cases} V_{REFP} - V_{REFN}, & V = +1 \\ 0, & V = 0 \\ V_{REFN} - V_{REFP}, & V = -1 \end{cases} \quad (8)$$

When only considering reference voltage mismatch (ΔV), the behavior of the feedback voltage V_{FB} is identical to that in the ideal case ($V_{FB,ideal}$). When only considering capacitor mismatch (ΔC), the behavior of the feedback voltage V_{FB} can be represented as:

$$V_{FB,\Delta C} = \begin{cases} (V_{REFP} - V_{REFN}) + V_{REFP} \Delta C, & V = +1 \\ 0, & V = 0 \\ (V_{REFN} - V_{REFP}) - V_{REFP} \Delta C, & V = -1 \end{cases} \quad (9)$$

According to Equation (9), if only capacitor mismatch exists, the feedback DAC remains absolutely linear. When both capacitor and reference voltage mismatches ($\Delta C \Delta V$) occur simultaneously, the behavior of V_{FB} can be represented as:

$$V_{FB,\Delta C + \Delta V} = \begin{cases} V_{FB,\Delta C} - \Delta V \Delta C, & V = +1 \\ 0, & V = 0 \\ V_{FB,\Delta C} - \Delta V \Delta C, & V = -1 \end{cases} \quad (10)$$

From Equation (10), it shows that the feedback DAC introduces a small nonlinearity term ($\Delta C \Delta V$). Fortunately, this error term primarily manifests as even-order harmonics, which can be suppressed by a fully differential system. Therefore, the tri-level feedback scheme has an advantage in fully differential systems compared to pseudo-differential systems. In target applications, the tri-level feedback DAC is still considered absolutely linear, eliminating the need for additional mismatch error calibration circuits. This makes the 1.5-bit quantizer scheme more attractive for energy-efficient applications and mature processes. Additionally, it relaxes the noise requirements for the

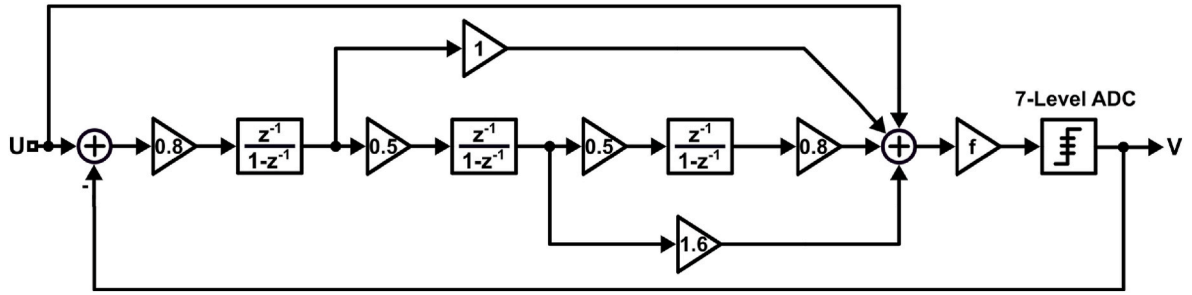


Fig. 9. The system block diagram of third-order 7-level CIFF DSM.

feedback reference voltage. Compared to a 1-bit quantizer, it is better suited for the design of high-precision, energy-efficient delta-sigma modulators.

4. The potential of using quantizers to optimize DSM system design

1-bit, 1.5-bit, and multi-bit quantizers exhibit different behavioral characteristics in DSM systems. Using the previous method, the equivalent gain k of the quantizer can be well captured, significantly facilitating DSM system design.

To avoid the additional overhead of mismatch noise shaping, the 1-bit quantizer becomes the optimal choice. The k of the 1-bit quantizer increases as the input decreases, which is beneficial. However, the higher quantization noise makes it less suitable for high dynamic range applications.

Multi-bit quantizers have lower quantization noise, which allows the system to achieve a larger MSA, making them more suitable for high dynamic range applications. On the other hand, the equivalent gain of multi-bit quantizers remains almost constant and is typically around 1, simplifying the design of DSM loop filters. However, multi-bit quantizers require additional mismatch error shaping, which introduces complexity and is not ideal for low-power designs, especially when using mature manufacturing processes.

The 1.5-bit quantizer demonstrates more balanced performance in DSM systems. For large input amplitudes, the k of a 1.5-bit quantizer with tri-levels is similar to that of a 1-bit quantizer, but it has smaller quantization error. Therefore, the 1.5-bit quantizer can achieve a higher peak SQNR and MSA. In contrast to multi-bit quantizers, the tri-level feedback scheme, based on reusing the input sampling capacitors from the first-stage integrator, is linear, eliminating the additional overhead of mismatch noise shaping. Therefore, the 1.5-bit quantizer is more suitable for high-precision, energy-efficient DSM designs using mature processes. However, the 1.5-bit quantizer has two drawbacks. The first is its relatively poor linearity, as the equivalent gain k is highly dependent on the quantization state. Although the quantizer's nonlinearity is suppressed

by the loop filter, it still needs to be considered in practical system designs. Fortunately, the linearity of the 1.5-bit quantizer is strongly correlated with V_{TH} , and simulations using the DSM system model can be used to find the V_{TH} value that minimizes distortion. The second drawback is that, at small input amplitudes, the equivalent gain k of the 1.5-bit quantizer experiences a sudden drop. This causes the system's loop gain to decrease abruptly, making it difficult for the DSM to resolve small amplitude signals. As a result, DSM based on the 1.5-bit quantizer have a relatively low dynamic range. The extent to which the dynamic range is affected increases with the proportion of quantization noise within the system's bandwidth.

Despite the challenges posed by the gain of the quantizer in DSM design, it also injects more vitality into the design process. By programming the gain of the quantizer, we can achieve variable loop gain in the DSM, making the system smarter. For example, for large input am-

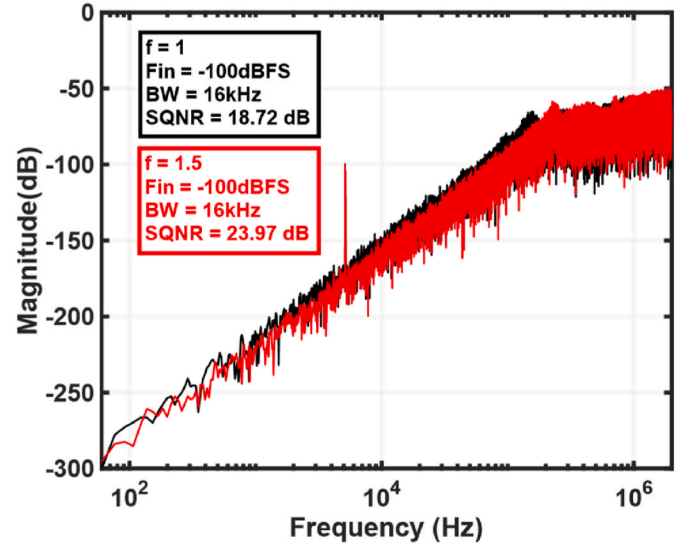


Fig. 10. Simulated the output spectrum of the DSM under different threshold voltage scaling factors.

plitudes, reducing the effective gain of the quantizer can decrease the loop gain of the DSM, allowing for a larger MSA. Conversely, for small input amplitudes, increasing the effective gain of the quantizer enhances the DSM's loop gain, resulting in stronger noise-shaping capabilities. A simple way to dynamically adjust the quantizer's effective gain is by programming the quantizer's threshold voltage. Additionally, previous work has demonstrated that the dynamic range limitation of a 1.5-bit quantizer can also be improved by dynamically scaling the quantizer's threshold voltage [18]. Assuming that the effective gains k_1 and k_2 are achieved by dynamically adjusting the quantizer's threshold voltage V_{TH} , the In-band quantization noise (IBQN) gain obtained through dynamic adjustment of V_{TH} can be calculated as:

$$\Delta IBQN \approx 10 \log_{10} \left(\frac{k_2}{k_1} \right)^2 \quad (11)$$

An optimized example of a low-distortion third-order 7-level CIFF DSM is shown in Fig. 9. Assuming the system has a sampling frequency of 4 MHz, a bandwidth of 16 kHz, and an input frequency of approximately 5.127 kHz. The DSM's SQNR curve under a -100dBFS is depicted in Fig. 10. When the threshold voltage scaling factor f of the 7-Level quantizer changes from 1 to 1.5, the SQNR improvement is approximately 5.3 dB, which is quite significant. The improvement increases as the bandwidth expands. It is worth noting that V_{TH} scaling can be easily implemented in a 7-level SAR ADC with only an additional scaling capacitor [18]. The monitor can be implemented using a Schmitt digital comparator along with reusing the decimation filter following the DSM. The errors introduced by delays or switching are suppressed by the loop filter and can be considered negligible. The power consumption

of this part is almost negligible for the DSM.

Furthermore, this optimization method can also be applied to real-time dynamic optimization. Although sudden changes in the quantizer threshold voltage may introduce some transient disturbances, these disturbances are largely suppressed by the loop filter and can be considered negligible. Additionally, more complex systems, such as hybrid architectures, can also benefit from design optimizations based on the quantizer.

5. Conclusion

This article presents a system design optimization method for discrete-time DSMs. The behavior-level modeling of quantizers is discussed in detail, including the empirical prediction method for equivalent gain, characteristics of mid-tread and mid-rise quantizers, as well as the relationship between input amplitude, threshold voltage, and equivalent gain. The behavior of the quantizer in a DSM system is also examined, covering the linear model characteristics of the quantizer in DSM, nonlinear considerations in DSM, limitations of empirical evaluation methods, and linearity considerations of feedback DACs. Furthermore, the approach of optimizing the DSM based on the quantizer's threshold voltage is explored, along with future development trends. This development approach will guide the design and optimization of DSM systems.

CRedit authorship contribution statement

Ziqiang Peng: Conceptualization, Methodology, Software. **Cong Wei:** Data curation, Writing – original draft. **Lijie Huang:** Conceptualization, Methodology. **Jinze Lai:** Data curation. **Xiaoqiang Lu:** Visualization, Investigation. **Rongshan Wei:** Reviewing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

No data was used for the research described in the article.

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